

Gains from Exchange

Principles of Microeconomics

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Nothing New Was Made. Both Walked Away Richer. How?

Sam hands Joe some granola bars. Joe hands Sam some bottles of water. No bar was baked and no bottle was filled.

Both say they are better off than before.

If trading only moves existing things around, where did the gain come from?

Last Time: How Much. Today: Why Both Sides Gain.

LAST TIME

How big is the response, and who cares?

Elasticity, revenue, and what makes demand more elastic.

TODAY

Why do both sides of a trade gain, and when does trading stop?

Personal worth, blocked trades, middlemen.

NEXT

Who coordinates millions of trades among strangers?

Market prices.

When Does the Next Trade Help Both People?

01

Trade Creates Wealth

Wealth is not stuff. The same goods are worth more in the hands that value them most.

Two Identical Parcels. A Newcomer Offers Each Man a Deal.

	The newcomer's deal
Sam	pays 7 bars for 10 bottles
Joe	gets 5 bars for 10 bottles
Newcomer	carries the water, keeps the difference

Every family starts with 20 bottles of water and 20 granola bars. The newcomer has no parcel. Assume arranging this fixed ten-bottle deal uses no resources valuable elsewhere.

Two Identical Parcels. A Newcomer Offers Each Man a Deal.

	The newcomer's deal	Would have agreed to
Sam	pays 7 bars for 10 bottles	up to 8 bars
Joe	gets 5 bars for 10 bottles	as few as 4 bars
Newcomer	carries the water, keeps the difference	—

Sam would have paid up to 8 bars and paid 7. Joe would have taken as few as 4 and got 5.
Arranging the deal has no opportunity cost. Who gains, and by how much?

Two Identical Parcels. A Newcomer Offers Each Man a Deal.

	The newcomer's deal	Would have agreed to	Ends with	Gain
Sam	pays 7 bars for 10 bottles	up to 8 bars	30 water, 13 bars	1 bar
Joe	gets 5 bars for 10 bottles	as few as 4 bars	10 water, 25 bars	1 bar
Newcomer	carries the water, keeps the difference	—	2 bars	2 bars

Sam gains 1 bar, Joe 1, the newcomer 2: four bars of net gain, measured in bars. Water moves to Sam, who values this bundle more. The dealer keeps two existing bars.

The same goods leave all three better off.

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For this fixed deal, changing payments redistributes the gain. Agreeing to trade creates it.

02

Personal Worth and the Stopping Point

With costless exchange, trading closes the gap in marginal worth.

Both Expect to Gain When the Price Lies Between Their Worths

Personal worth your reservation value: the most you would give to acquire a good, or the least you would accept to surrender it.

TEST IT *Paying \$32 tells us the price. Does it reveal your maximum?*

You would pay up to \$40 for a used jacket. The seller would let it go for as little as \$25. It sells for \$32.

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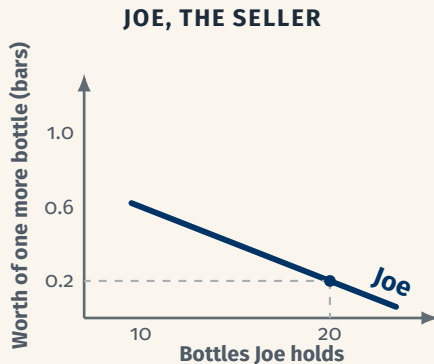
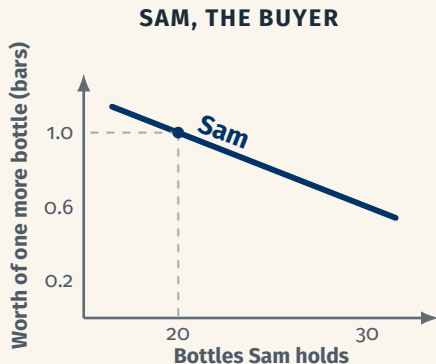
\$40 — \$32 = \$8 to the buyer

\$32 — \$25 = \$7 to the seller

Seller's minimum \$25 < price \$32 < buyer's maximum \$40.

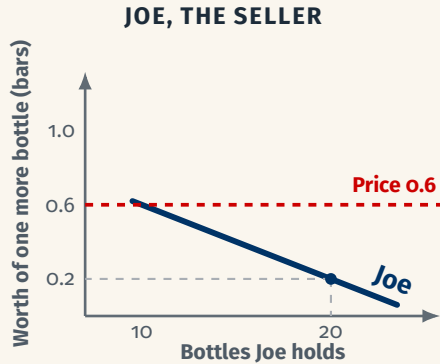
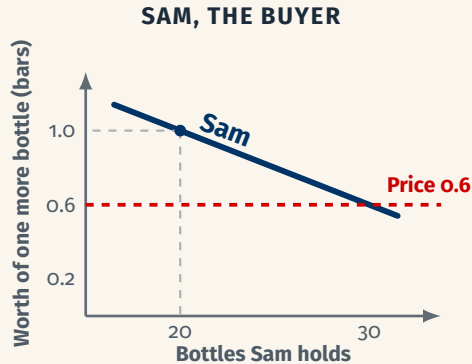
At this price, both expect to gain, before transaction costs.

Different Worths at the Same Start Make the Trade



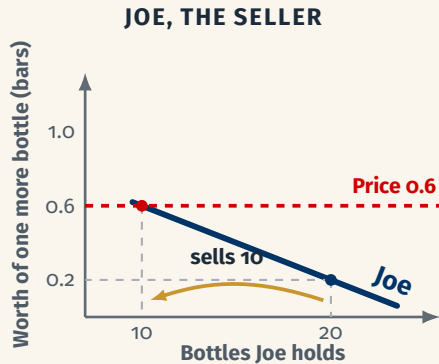
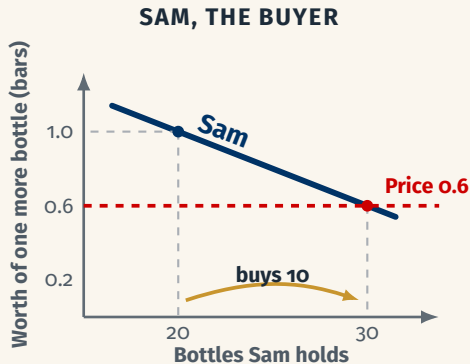
Both start with 20 bottles. One more is worth 1.0 bars to Sam and 0.2 to Joe. Now they trade directly, without a dealer or arranging costs; these schedules stay fixed.

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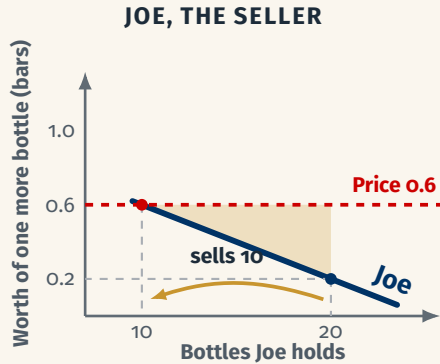
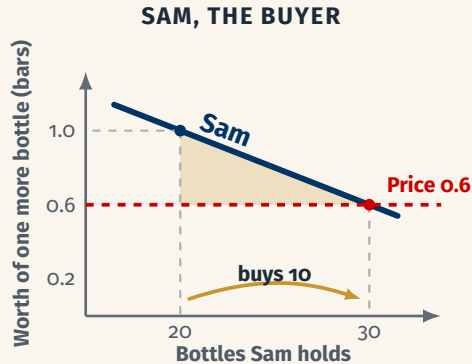
Direct exchange at a uniform 0.6 bars per bottle. Sam pays Joe directly. His worth exceeds the price; Joe's is lower.

Different Worths at the Same Start Make the Trade



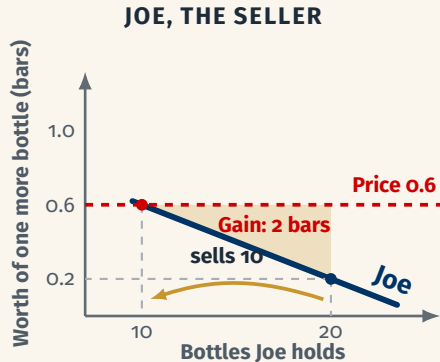
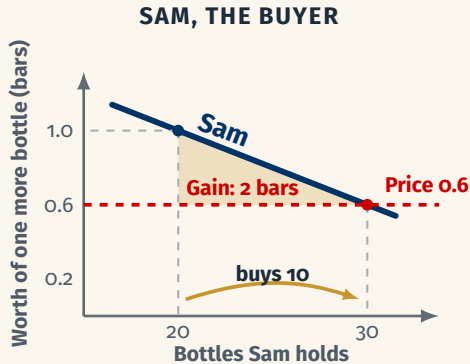
Sam buys from 20 up to 30 as the worth of each added bottle slides down to 0.6; **Joe sells from 20 down to 10** as each remaining bottle's worth climbs to 0.6. Then they stop.

Different Worths at the Same Start Make the Trade



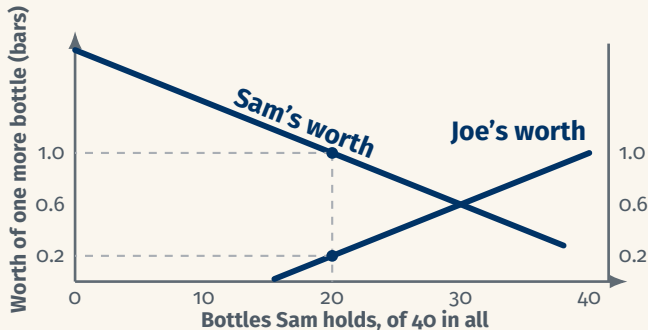
The shaded areas are the gains from direct exchange. Why does Joe gain even though he ends with fewer bottles?

Different Worths at the Same Start Make the Trade



The same four-bar gross gain; a different division. Sam pays Joe 6 bars for ten bottles. Each gains 2; no dealer takes a cut.

With Costless Exchange, Trade Stops Where the Worths Meet



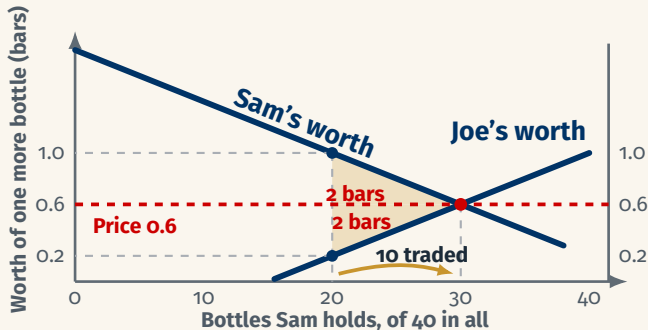
DIRECT EXCHANGE

Sam holds the number on the axis

Joe holds the rest of the 40.

The same two lines on one frame. Joe's bottles count from the right edge, so his falling worth line appears to rise. Both start at 20, worths 1.0 and 0.2.

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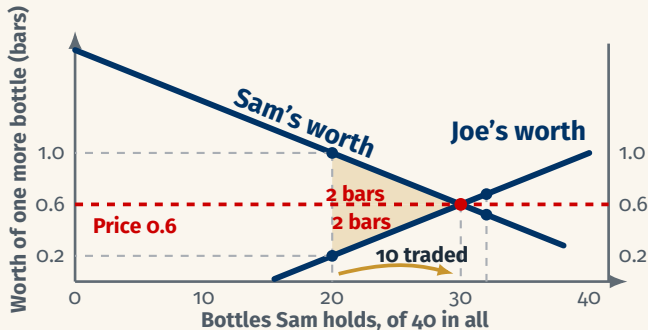
AT THE CROSSING

Sam 30, Joe 10, price 0.6

2 bars of gain to each.

Each bottle traded slides Sam down his line and Joe up his own, and the bowtie fills until the lines cross at 0.6.

With Costless Exchange, Trade Stops Where the Worths Meet



AT THE CROSSING

Sam 30, Joe 10, price 0.6

2 bars of gain to each.

IF THEY OVERSHOOT

Reversing helps both

Joe's worth: 0.68. Sam's: 0.52.

At the crossing, no swap can help one man without hurting the other. Economists call that allocation Pareto-optimal.

Pareto-Optimal Means No Gain Left. It Does Not Mean Fair.

Pareto-optimal allocation no feasible reallocation can help one person without hurting another. Costless trade gets there when rights are recognized, choices are informed and voluntary, and no unpriced effect falls on outsiders.

TEST IT *Sam ends with 30 bottles, Joe with 10. Fair?*

IT IS NOT

Productive efficiency

That test is about waste in production, not about who holds what afterward.

IT DOES NOT SAY

That the result is fair

The criterion takes initial holdings as given. It does not judge whether they were just.

The “is”: no mutual gain is left. The “should” is still yours.

03

Blocked Trades and Middlemen

Forbid a trade and the value moves. Arrange one cheaply and the value grows.

The Swap Is Forbidden. Does the Value Disappear?

You get a parking permit; a friend gets a desk in the library stacks. Each would rather have the other's. The swap is forbidden.

Why would the university forbid it?

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ALLOCATION POWER

No cash receipt

The university bans transfers. An administrator still assigns scarce spaces.

CAPTURED

Favors, goodwill, discretion

Whoever controls the space may capture benefits in permitted forms.

USED UP

Queues, lobbying, paperwork

Nonprice competition burns real resources; part of the value is dissipated, not captured.

Blocking the trade kills the gain, not the value of control.

“Cut Out the Middleman.” And Do Whose Work?

A friend says: buy your groceries straight from the farmer and cut out the middleman. You will save money.

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FIND

Who has what you want, and wants what you have

The camp's good trades waited for the newcomer who arranged them.

JUDGE

Quality and reliable delivery

Observable cheating threatens future business, making an honest reputation valuable.

CARRY

Travel, storage, timing

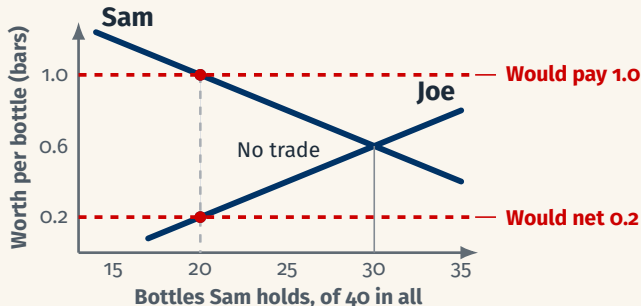
Food from farmers, shoes from the factory, milk from the dairy: a life of shopping.

The work does not vanish. It moves to you.

Sometimes you are the cheaper arranger; then do it yourself.

Lower Transaction Costs Make More Trades Worthwhile

Now each bottle traded uses real resources. Closer view of the same worth curves.



PREDICT THE TRADE

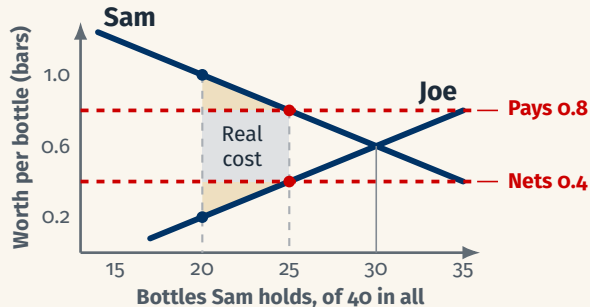
Cost / bottle	Bottles traded
0.8	0
0.4	?
0.2	?
0	?

Divisible water, measured in bottle equivalents.

The next trade's gain must exceed its real cost. At 0.8 bars per bottle, no positive amount pays. What if the cost falls?

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PREDICT THE TRADE

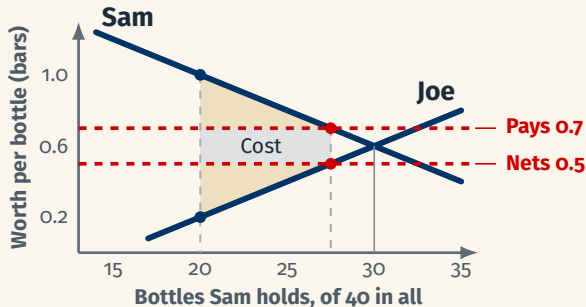
Cost / bottle	Bottles traded
0.8	0
0.4	5
0.2	?
0	?

Divisible water, measured in bottle equivalents.

At a cost of 0.4, five bottles move. The last marginal worth gap just covers arranging cost. Gray shows resources used.

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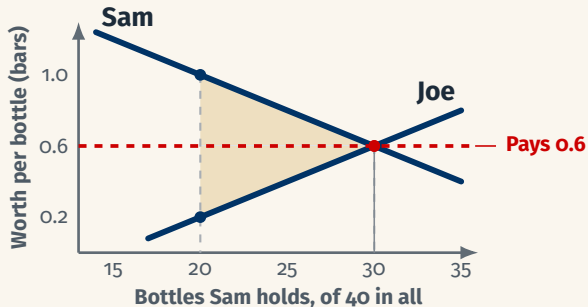
Cost / bottle	Bottles traded
0.8	0
0.4	5
0.2	7.5
0	?

Divisible water, measured in bottle equivalents.

At a cost of 0.2, 7.5 bottles move. These are real costs per bottle, unlike the dealer's fee for the fixed bundle.

Lower Transaction Costs Make More Trades Worthwhile

Now each bottle traded uses real resources. Closer view of the same worth curves.



PREDICT THE TRADE

Cost / bottle	Bottles traded
0.8	0
0.4	5
0.2	7.5
0	10

Divisible water, measured in bottle equivalents.

With no transaction cost, the worths meet: ten bottles move. A fee pays a middleman; only resources used are real costs.

A Lone Middleman Can Take the Whole Gain. Rivals Trim It.

The newcomer is the only trader in camp. What is the most he can charge for arranging the ten-bottle deal?

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At most 4 bars, the gap between 8 and 4.

At the limit both are indifferent; a smaller fee leaves strict gains.

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ONE DEALER, THEN TWO

	Water	Granola
Sam, the newcomer alone	30	13
Sam, with a rival	30	13.5
Joe, the newcomer alone	10	25
Joe, with a rival	10	25.5

THE SPREAD

2 bars per ten bottles → 1 bar

The rival offers Sam 6.5 and Joe 5.5. Each gains another half bar.

Rival offers and the option to trade directly limit the fee.

License Only “Approved” Traders. Who Gains?

The dealers propose a rule: only licensed, trained, ethical middlemen may trade in camp, and the existing dealers decide who qualifies.

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RESTRICTED ENTRY

A margin may be protected

If incumbents exclude capable rivals, customers lose alternative offers.

LICENSE PRICE

Future earnings have value today

A credible, durable right to future net earnings can command a price today.

QUALITY ASSURANCE

Screening may help buyers

Better quality can coexist with a protected spread. Compare benefits and costs.

What evidence would show which effect matters more?

The Next Trade Must Cover Its Real Cost

A buyer values a used lamp at \$18. Its owner would accept \$10. Arranging the trade uses resources worth \$9.

Can this trade leave both better off after arranging costs? What changes if an app reduces the real cost to \$3?

NEXT: MARKETS Millions of trades among strangers. How do prices coordinate them?

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Can this trade leave both better off after arranging costs? What changes if an app reduces the real cost to \$3?

The gross gain is \$8.

A \$9 real cost blocks the trade. At a \$3 real cost, \$5 remains to share. The app's fee determines payments; its resources used determine real cost.

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The Next Trade Must Cover Its Real Cost

WORTH AND GAINS

Exchange can improve both positions

Differing personal worths create room for gain. Payments divide it; the quantity of goods need not grow.

THE STOPPING RULE

The next gain must cover the next real cost

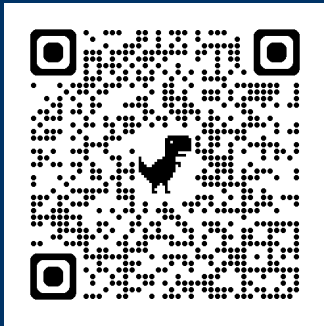
With zero transaction cost, worths meet. Middlemen can lower costs; entry and reputation discipline their service.

Bans can redirect or dissipate scarcity rents. Licensing can screen quality and restrict entry; evidence must distinguish the effects.

NEXT: MARKETS **Millions of trades among strangers. How do prices coordinate them?**

BEFORE YOU GO

Tell me what worked and what didn't.



Scan for a one-minute anonymous form.